

1 Benchmarking sequence, embedding and structure 2 based methods for eukaryotic virus protein annotation

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10 Outline (to delete)

- 11 • general overview of data and structures
 - 12 – TODO: add annotation of BFVD (eukaryotic vs prokaryotic viruses)
- 13 • performance of methods in hits
- 14 • comparison of methods in shared hits
- 15 • comparison of functional annotation per method to original
- 16 • how many are informative annotations
 - 17 – TODO: which conversions from hypothetical protein to informative (phages?)
- 18 • TODO: computational benchmark

19 Abstract

20 Introduction

21 The advent of protein language models (pLMs) and structural prediction promises new avenues
22 for protein annotation through structure homology. Here, we benchmarked four recent tools to
23 leverage these new techniques for protein annotation of eukaryotic viruses.

Results & Discussion

Dataset overview

Our test set of protein sequences consisted of 1,054 different viral entries taken at random from ICTV's VMR data (n=1,000 distinct species). These test dataset viruses covered all eukaryotic viral Kingdoms, with the majority of proteins originating from *Orthornavirae* (RNA viruses) and *Bamfordvirae* (dsDNA viruses) (see Figure 1A), despite the latter only representing 43 viruses (see Supplemental Figure 1).

Structure prediction of all 11,467 proteins from our test set with Boltz-2 resulted in high-quality predicted structures, with a median pLDDT score of 71.3 and 52.2% of all predicted structures having a pLDDT score above 70 (see Figure 1B). This shows that the structure prediction with default settings of Boltz-2 results in reliable structure predictions without the need for extensive fine-tuning of parameters or MSA generation strategies.

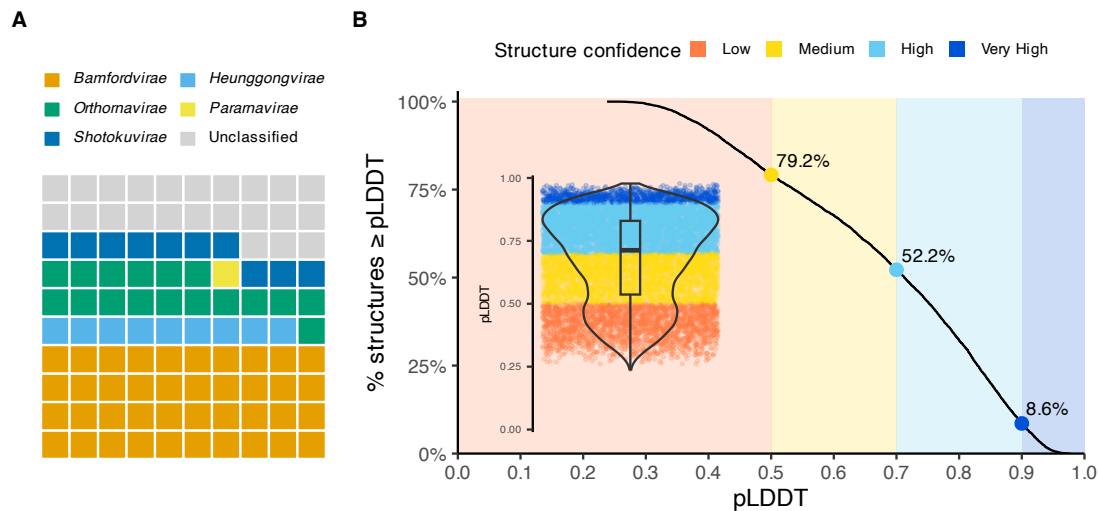


Figure 1: A) Waffle plot showing the distribution of the number of proteins from the different viral Kingdoms. Each tile roughly corresponds to 110 proteins. B) A cumulative distribution function plot showing the percentage of predicted structures from the test set with a pLDDT score above the pLDDT score on the x-axis. The inset shows a violin- and boxplot with all pLDDT scores from the predicted structures (n=11,467).

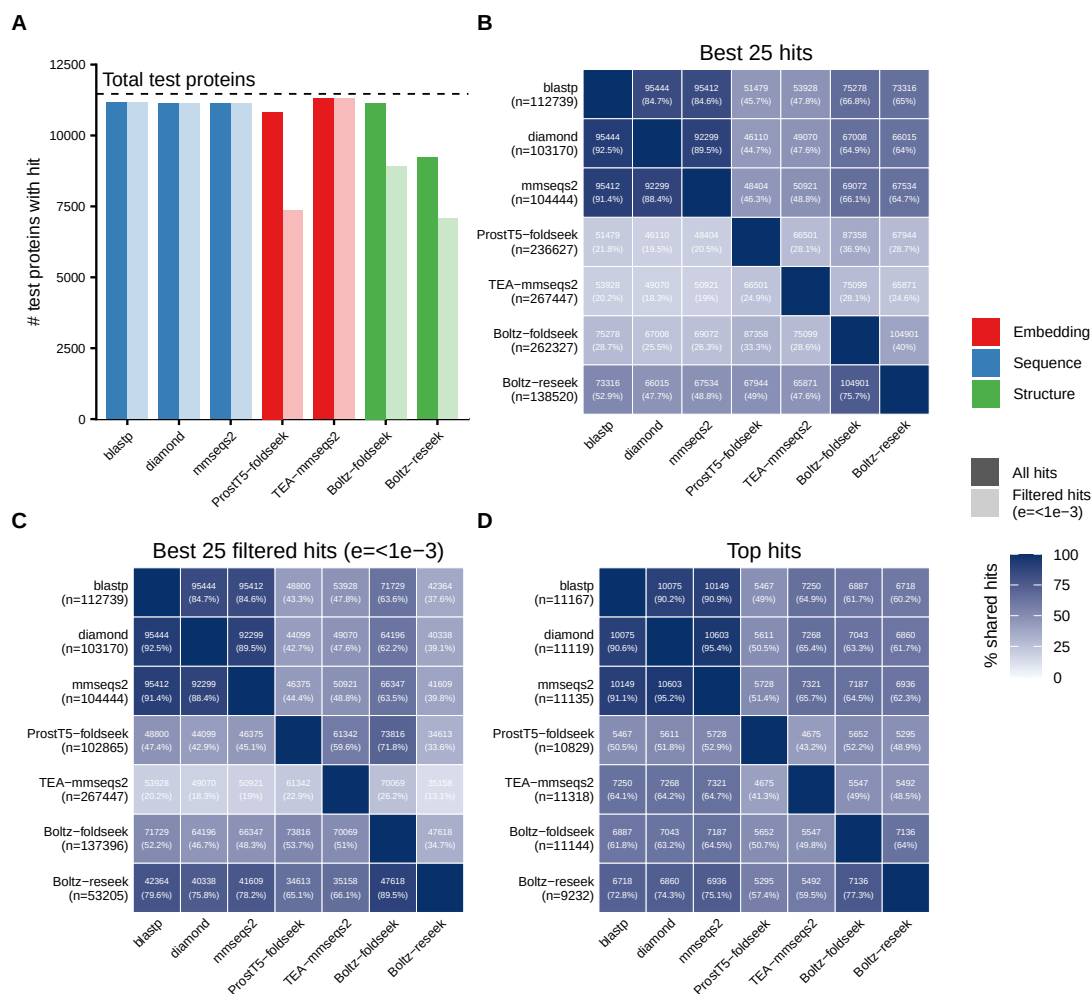
Performance of the different methods

We started by evaluating how many of the proteins in our test set got a hit against the BFVD for each method (see Figure 2A). In addition, we also filtered the results of each method on an e-value lower than 1e-3 to see how this affected the number of hits for each test protein. The methods relying on structure homology clearly had a lower number of test proteins with hits

41 against the BFVD compared to the sequence homology-based methods. In contrast, TEA had
42 the most number of test proteins with a hit, almost reaching the total number of test proteins
43 ($n=11,318$). The three sequence based methods had a comparable number of test proteins
44 with a hit (blastp=11,165; mmseqs=11,135 and diamond=11,119). After filtering on e-value
45 lower than $1e-3$, the number of proteins with hits decreased only for the structural methods
46 on average by 25%. This suggests that sequence based methods are still more sensitive in
47 detecting homology for eukaryotic viruses than structure based methods, despite the promising
48 results for bacteriophages¹. As the quality of our structure predictions was rather high (see
49 Figure 1B), this lower sensitivity is likely not due to poor structure predictions.

50 When comparing the BFVD hits yielded by all methods against each other (see Figure 2B-D),
51 we observed as expected that the sequence homology-based methods had a high overlap in
52 shared hits. This was true when comparing the best 25 hits for each test protein, the best 25 hits
53 afterwards filtered on e-value and only the top hits. In contrast, the embedding and structure
54 homology based methods had low overlap (~60%) in resulting hits with both the sequence
55 based methods as each other. The percentage of overlapping hits did increase when only
56 looking at the top hits but stayed well below the percentage of overlap between the sequence
57 based methods. It is worth noting that Reseek seemed to perform worse in this metric when
58 looking at the filtered results (see Figure 2C), however this is due to the large reduction in
59 number of hits after filtering. This reduction is due to the more stringent e-value calculations in
60 Reseek compared to Foldseek^{2,3}. It is worth noting that while ProstT5 was the worst performer
61 across the three comparisons, TEA performed better in the top hits comparison and was on
62 par with the structure based methods when comparing to the sequence based methods.

63 Overall, the embedding and structure based methods, that should be better to detect remote
64 homologs, had low agreement on shared hits among each other. And although they did find
65 more hits for the test proteins on default settings (see y-axis Figure 2B), this was almost
66 entirely removed after filtering the hits on e-value (see y-axis Figure 2C). This effect was most
67 pronounced for Reseek, which claims to calculate the most correct e-values² and thus leaving
68 the question if the Foldseek searches do not report too many false positives.



Functional annotation comparison

To functionally annotate our test proteins for each method, weights in form of the negative log e-value (or a weight of 1000 in case of an e-value of 0) were given to each hit. These weights were summed per functional category (assigned to each entry in the BFVD; see methods) and the category with the largest weight was chosen as the functional category for each test protein. Informative categories were prioritized over uninformative categories (ie. “Hypothetical Protein”, “unknown”, “Other function”) when the weight of the informative category was at least 50% the weight of the uninformative categories.

77 To evaluate the accuracy of the benchmarked methods, we considered the test proteins with an
 78 informative annotation (n=2,938) (see Figure 3A) as the ground truth to compare our results to.
 79 All methods improved the number of informative categories for all test proteins (see Figure 3A),
 80 with mostly the embedding and structure based methods seeming to improve the number of
 81 informative annotations. However, when comparing the functional annotations to the “ground
 82 truth”, the embedding and structure based methods underperformed compared to the sequence
 83 based methods (~67% and ~76% vs ~83% agreement with ground truth; see Figure 3B).

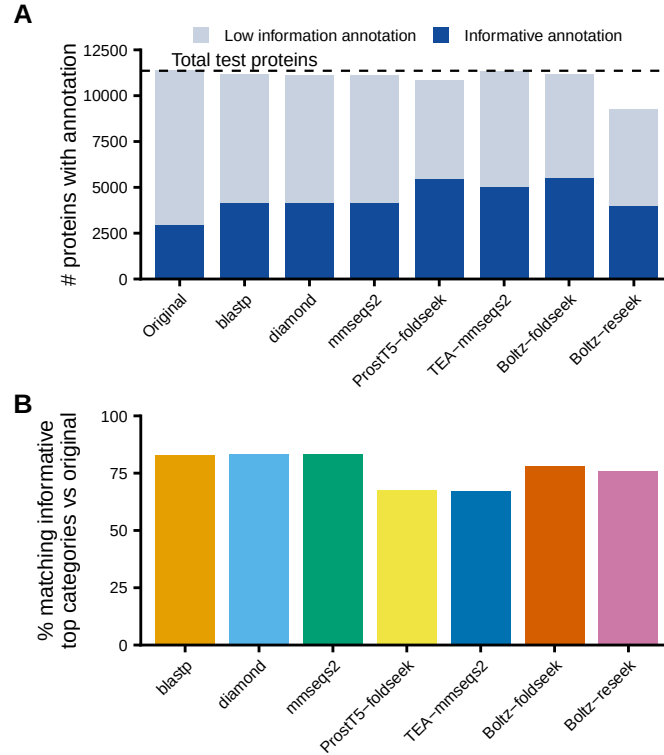


Figure 3: A) Bar plot showing the number of proteins that received an informative functional annotation (i.e. not “Hypothetical Protein”, “unknown” or “Other Function”) for the original protein descriptions and after applying each method. B) Barplot showing the percentage of queries that have a matching functional category for each method compared to the informative original functional categories (regarded as the ground truth).

84 Computational benchmarking

85 Conclusion

86 Although new methods like pLMs and structure homology searches have shown great promise in
 87 detecting remote homologs^{2,4,5} including bacteriophage protein searches¹, this does not seem
 88 to be the case for eukaryotic virus proteins. If we consider that structural homology searches
 89 are most useful to find homologs with a clearly defined protein structure, a possible explanation
 90 could be that most structural proteins (eg. capsid proteins) are already well annotated in

sequence databases compared to more globular, complex proteins, and therefore the advantage of using structural homology searches in eukaryotic virus protein annotation might be limited.

Material and Methods

Dataset selection

ADD SOFTWARE VERSIONS

Test set

To benchmark different methods for eukaryotic virus protein annotation, we selected 10% of the entries in ICTV's Virus Metadata Resource (VMR) v40.2 that did not have bacteria or archaea as designated host (n=1,054; 1,000 distinct species). We subsequently extracted their GenBank accession numbers (n=1,402) and downloaded all associated proteins (n=11,360).

Protein sequences that were longer than 2,500 amino acids were split into multiple chunks of maximum 2,500 amino acids so that structure prediction with Boltz-2⁶ would not run out of GPU memory (see below). This resulted in a final test set of 11,467 protein sequences from 1,000 eukaryotic virus species.

Database

The Big Fantastic Virus Database (BFVD)⁷ was used as the reference database for both sequence and structure homology-based annotation. The BFVD is a comprehensive database of high-quality predicted protein structures from the viral sequence representatives of the UniRef30 clusters. UniRef30 is a UniProt database that clusters protein sequences at 30% sequence identity and selects a representative sequence for each cluster, these representatives thus have a UniProt accession. Since a substantial amount of BFVD's included proteins are no longer available in UniProtKB as they are not part of reference proteomes, we converted their UniProt accessions to UniParc IDs.

Functional categories for each BFVD entry were derived from protein names associated with the corresponding UniParc IDs (could be from UniProtKB/TrEMBL, RefSeq and/or GenBank) for eukaryotic viruses (also see below) or by classifying them into PHROG-like⁸ categories with Empathi⁹ in the case of bacteriophage proteins. The distinction between eukaryotic virus and bacteriophage proteins was made based on the TaxID information from the UniParc protein ID.

The BFVD database was further converted into the required formats the search tools used in this benchmarking effort (see sequence and structure homology-based annotation).

Functional classifier

Functional categories for eukaryotic virus proteins were designed with Perplexity AI in deep research mode using the following prompt: "Can you give me an exhaustive list of common

124 protein functions in eukaryotic viruses? Use <https://viralzone.expasy.org> and <https://ictv.global/report/genome> as sources. Give me a machine readable file with all these functional
125 categories.” The resulting functional categories were further manually curated to ensure that
126 they were mutually exclusive. The final list of functional categories is available in **Supplementary**
127 **Table 1**.

129 A custom python script was used to assign functional categories to BFVD proteins based on
130 pattern matching. Perplexity AI was used to assign the most common patterns in all BFVD
131 protein names from UniParc to the most appropriate functional category. The pattern matching
132 rules, functional categories, and classification script were subsequently refined through manual
133 curation to optimize the functional assignment based on protein names. If no keywords in the
134 protein match, the protein is assigned to the “Hypothetical Protein” category.

135 The same functional classifier script was also used on the protein names from our test set
136 to assign functional categories to the eukaryotic virus proteins downloaded from the ICTV
137 exemplar species.

138 **Sequence homology-based annotation**

139 For sequence homology annotation, we used three state-of-the-art tools widely used in func-
140 tional annotation pipelines and sequence similarity searches. As gold standard, blastp¹⁰ was
141 used with an e-value cutoff of 1e-3 to align the test set amino acid sequences against the amino
142 acid sequences of the BFVD entries. In addition, we used diamond blastp¹¹ in sensitive mode
143 and mmseqs2 easy-search¹² with default settings; both have a default e-value cutoff of 1e-3.

144 **Protein embedding-based annotation**

145 **ProstT5**

146 ProstT5¹³ is a pLM, based on ProtT5-XL-U50¹⁴, finetuned for translating amino acid sequences
147 into structure representations with 3Di sequences. We used ProstT5 within foldseek⁴ using
148 default settings to generate 3Di sequences for our test set and search them against the BFVD.

149 **TEA**

150 In contrast to ProstT5, The Embedded Alphabet (TEA)⁵ uses the ESM2 pLM¹⁵ to generate
151 embeddings for protein sequences and converts these embeddings into TEA sequences using
152 this novel structural alphabet. Despite TEA and 3Di both consisting of 20 characters, TEA
153 is different from 3Di in terms of the numbers and distribution these TEA characters map to
154 secondary structural elements, promising a better representation of different structural elements
155 across multiple alphabet characters.

156 We converted both our test set and the BFVD into TEA sequences using the `tea_convert`
157 pipeline with default settings, and subsequently performed sequence similarity searches using

158 mmseqs2 easy-search¹² with a custom substitution matrix (matcha substitution matrix provided
159 with TEA) and gap scoring settings (--comp-bias-corr 0 --mask 0 --gap-open 18 --gap-
160 extend 3) to search the test set against the BFVD.

161 **Structure homology-based annotation**

162 **Structure prediction**

163 For the structure prediction of our test set proteins, we relied on Boltz-2⁶ (default settings) with
164 multiple sequence alignments (MSAs) generated by a local, adapted ColabFold installation¹⁶
165 (see <https://github.com/gbouras13/colabfoldv>). To increase the depth and diversity of the MSAs,
166 we used the default ColabFold sequence databases, supplemented with extra *Orthornavirae*
167 sequences¹⁷ and viral sequences used in the structure prediction for the Phold database¹.
168 Structures were predicted on an NVIDIA H100 GPU with 80GB of VRAM.

169 **Foldseek**

170 Foldseek⁴ converts predicted protein structures into 3Di sequences and subsequently performs
171 fast structure similarity searches using these 3Di sequences. Here, we used foldseek easy-
172 search with default settings to search the predicted structures of our test set against the
173 BFVD.

174 **Reseek**

175 Reseek² aligns structures in a similar way to Foldseek, but uses a different structural alphabet
176 consisting of $\sim 10^{11}$ states, promising better sensitivity for remote homologs.

177 We used an adapted version of Reseek v2.7 in sensitive mode that switches the loading of the
178 queries and database to get more accurate e-value calculations (see [this discussion](#)).

179 **Computational benchmarking**

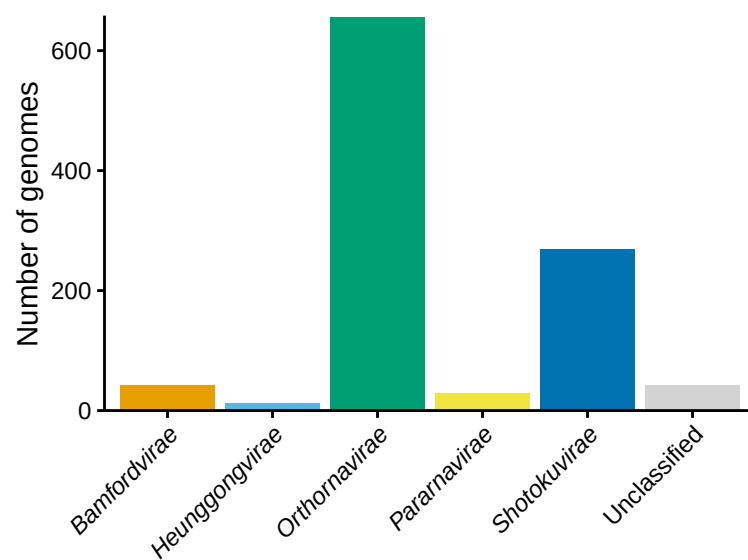
- 180 • hyperfine

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Supplementary Figures



Supplemental Figure 1: Bar plot showing the distribution of the number of genomes from the different viral Kingdoms in the test set.